
Use of Higher Order Schemes for Flow Computations

Internship Presentation

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Formal order of accuracy - change in the truncation error due to the increase in power (higher order).

General truncation error form is denoted as:

$$\epsilon_{\tau} = (\Delta x)^m \alpha_{m+1} + (\Delta x)^{m+1} \alpha_{m+2} + (\Delta x)^{m+2} \alpha_{m+3} + \dots + (\Delta x)^n \alpha_{n+1}$$

where α denotes higher order derivatives multiplied by some constant value

Discretization errors - rate at which the error decreases when the grid becomes finer.

$$\mathcal{L}(\phi) = \mathcal{L}_h(\phi) + \tau_h$$

$$\mathcal{L}_h(\epsilon^d_h) = -\tau_h$$

Maintaining higher order schemes at all nodes



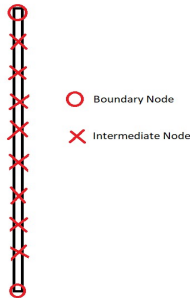
Let 'n' be the number of nodes, 'p' be order of the derivative to be computed 'q' be the required order of accuracy.

1. Biased schemes (Upwind/Downwind)

$$n = p + q$$

2. Central schemes

$$n = p + q - 1$$





Schemes and Error Analysis

1. FDS/BDS/CDS
2. Pentadiagonal and Tridiagonal Compact Schemes (Pade generalization)
3. Spectral schemes

Fourier Error Analysis

Generalized Pade scheme (7 points)

$$\beta \hat{f}_{i-2} + \alpha \hat{f}_{i-1} + \hat{f}_i + \alpha \hat{f}_{i+1} + \beta \hat{f}_{i+2} = c \frac{f_{i+3} - f_{i-3}}{6h} + b \frac{f_{i+2} - f_{i-2}}{4h} + a \frac{f_{i+1} - f_{i-1}}{2h}$$

$$f(x_i) = \sum_{q=-N/2}^{N/2-1} \hat{f}(k_q) e^{ik_q x}$$

$$\frac{df}{dx} = \sum_{q=-N/2}^{N/2-1} ik_q \hat{f}(k_q) e^{ik_q x}$$

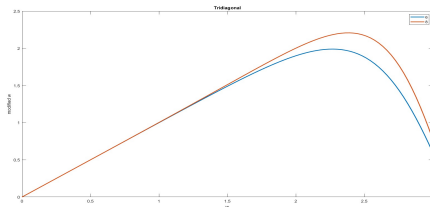
Note that dependent variables are assumed to be periodic for conducting the Fourier Analysis



$$\text{Modified wave no.} = \frac{a \sin(w) + (b/2) \sin(2w) + (c/3) \sin(3w)}{1 + 2\alpha \cos(w) + 2\beta \cos(2w)}$$

Tridiagonal Schemes

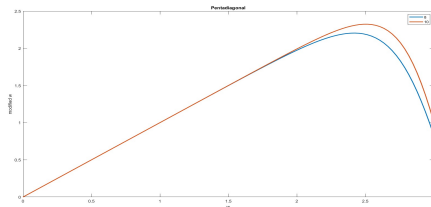
	a	b	c	α	β
6th Order	14/9	1/9	0	1/3	0
8th Order	19/12	1/6	0	3/8	0



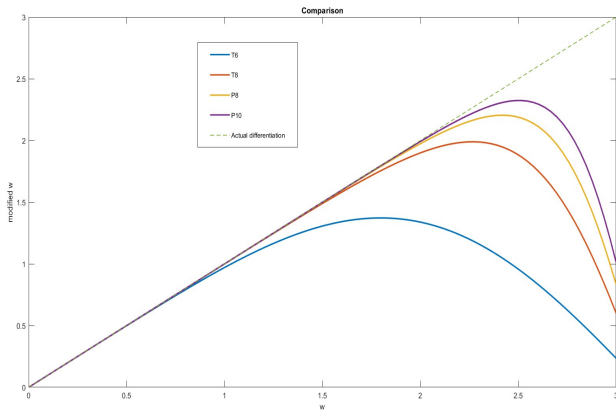


Pentadiagonal Schemes

	a	b	c	α	β
8th Order	40/27	25/54	0	4/9	1/36
10th Order	17/12	101/150	1/100	1/2	1/20



Comparison





Midpoint Interpolation

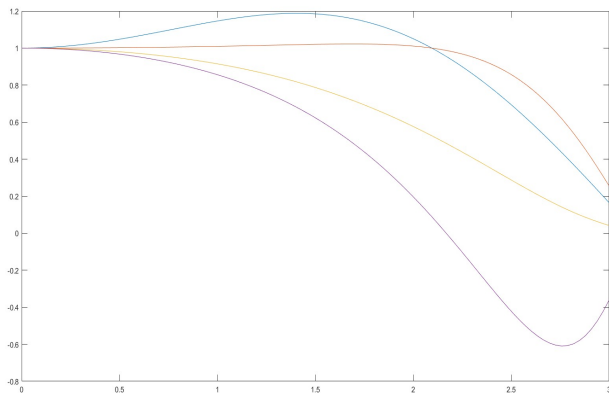
Half nodes: $x \pm 1/2, x \pm 3/2, x \pm 5/2$

$$\beta \hat{f}_{i-2} + \alpha \hat{f}_{i-1} + \hat{f}_i + \alpha \hat{f}_{i+1} + \beta \hat{f}_{i+2} = \\ \frac{c}{2}(f_{i+5/2} - f_{i-5/2}) + \frac{b}{2}(f_{i+3/2} - f_{i-3/2}) + \frac{a}{2}(f_{i+1} - f_{i-1})$$

Transfer function after inversion:

$$T = \frac{a \cos(w/2) + b \cos(3w/2) + c \cos(5w/2)}{1 + 2\alpha \cos(w) + 2\beta \cos(2w)}$$

Comparison





- 1 Ferziger, J.H. and Peric, M. (1999) Computational Methods for Fluid Dynamics. 2nd Edition, Springer-Verlag, Berlin.
- 2 Numerical analysis of spectral methods : theory and applications / David Gottlieb and Steven A. Orszag.
- 3 Compact Finite Difference Schemes with Spectral-like Resolution, Sanjiva. K .Lele.